

02

GUIDEBUDDY: AN INTELLIGENT ASSISTIVE SYSTEM FOR GUIDE DOG TEAMS

Wearable Design

Innovative Design

Blind-friendly

This project is a practice in intelligent systems design, aiming to enhance the safety and well-being of collaboration between visually impaired users and guide dogs through an integrated ecosystem of sensing, decision-making, and feedback.

Year: 2024 winter
Individual Project

CARING DESIGN

Guidebuddy

BACKGROUND

Guide dogs are essential companions for visually impaired individuals, providing navigation support and emotional companionship in daily life.

Challenges Faced by Guide Dogs

"UK Guide Dogs Association reports **74%** of working dogs display mental health concerns, with **18%** exhibiting weekly distress symptoms."



"Just **36%** of dog owners recognize canine mental health issues, with only 24% aware dogs can experience psychological distress."

Widespread bans on guide dogs in public spaces and transit systems severely limit blind individuals' mobility and compromise the wellbeing of assistance animals.



Physical Strain

Guide dogs endure significant physical and emotional stress, including leg health issues, particularly affecting their quality of life and working ability.



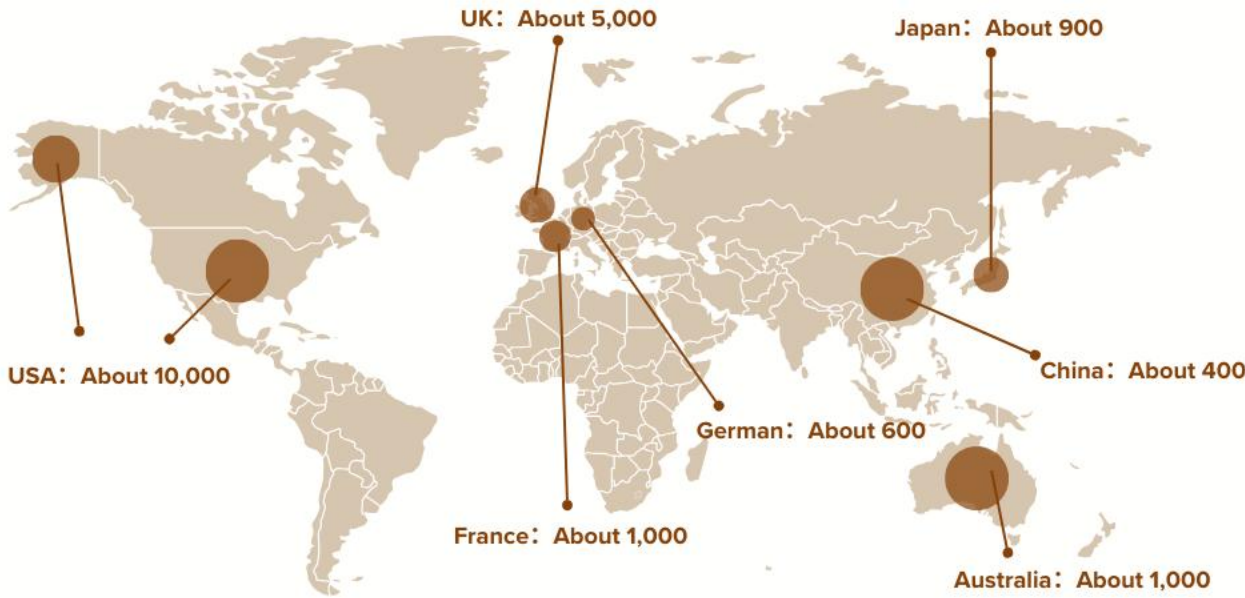
Technological Gap Analysis

Traditional harnesses lack biometric monitoring



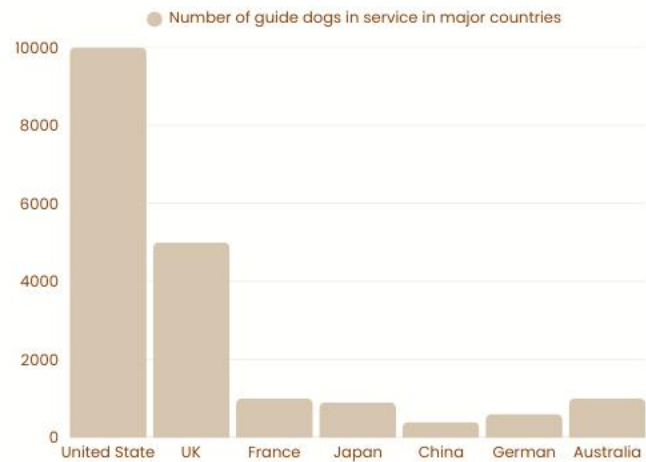
RESEARCH

How many guide dogs are there in the world?



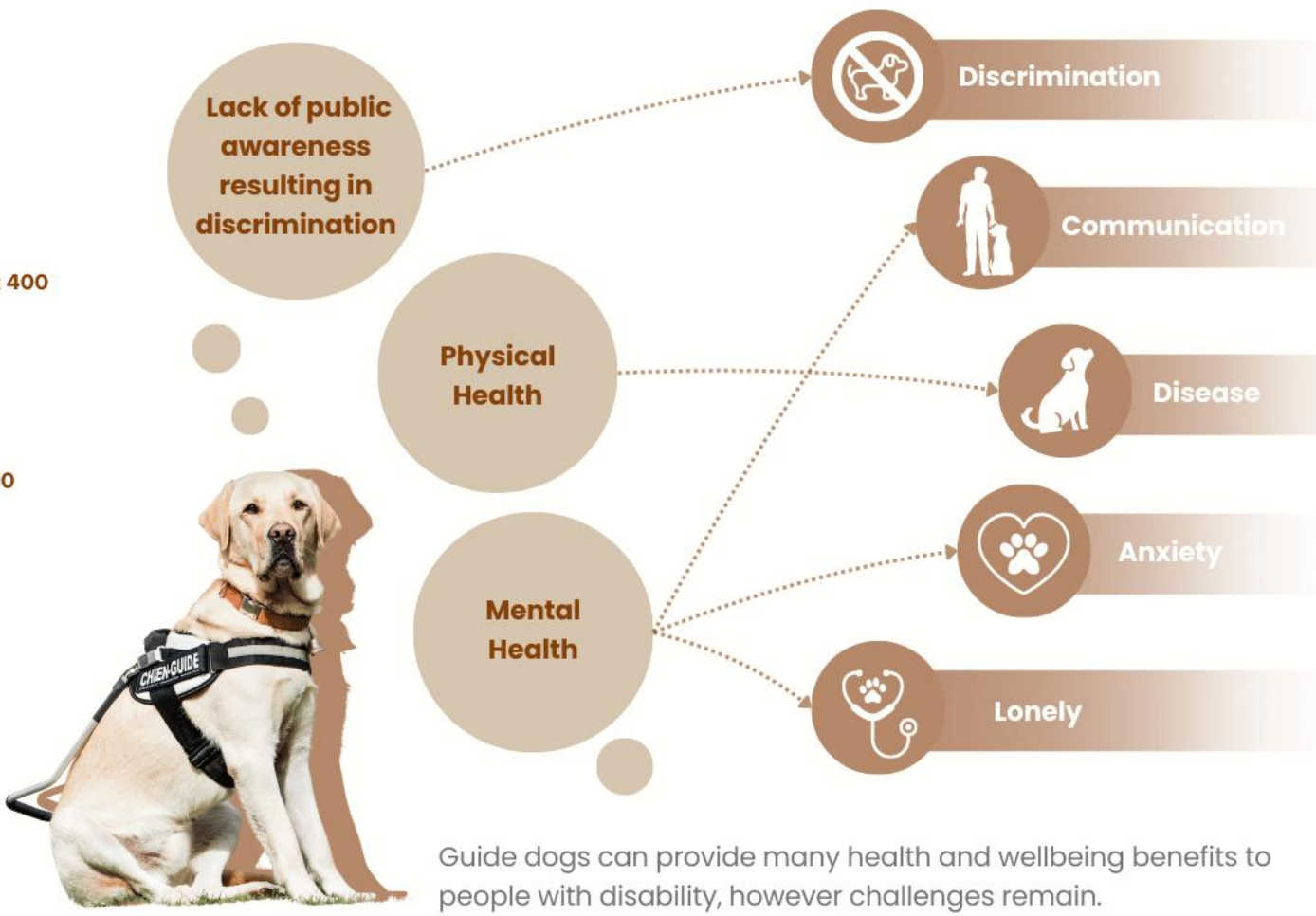
How many people in the world have a guide dog?

19,557 guide dogs are currently working worldwide. This means that **19,557** people who are blind or have low vision have the support of a guide dog to help them live life on their terms.



Problem Definition

How to use technology to improve the physical and mental health of guide dogs, while helping blind owners better understand and support the needs of guide dogs?



INTERVIEW

During interviews, both blind handlers and guide dog trainers were asked about their ability to perceive the dogs' emotional states in daily routines.

Paul

- 30/male/blind
- he was diagnosed with eye disease at the age of 16 called retinitis pigmentosa

"I can only sense extreme stress when my dog pants heavily or slows down suddenly."

Mrs. Li

- Principal of Guangzhou Guide Dog School

"Vibration alerts would help distinguish between normal fatigue and real distress."

INSIGHT

Through insightful analysis of the visually impaired's habits and guide dogs' working scenarios, we can gain a better understanding of the user context.

Guide dogs must walk, stand, and carry weight for long periods, leading to joint wear, muscle fatigue, and reduced working longevity.

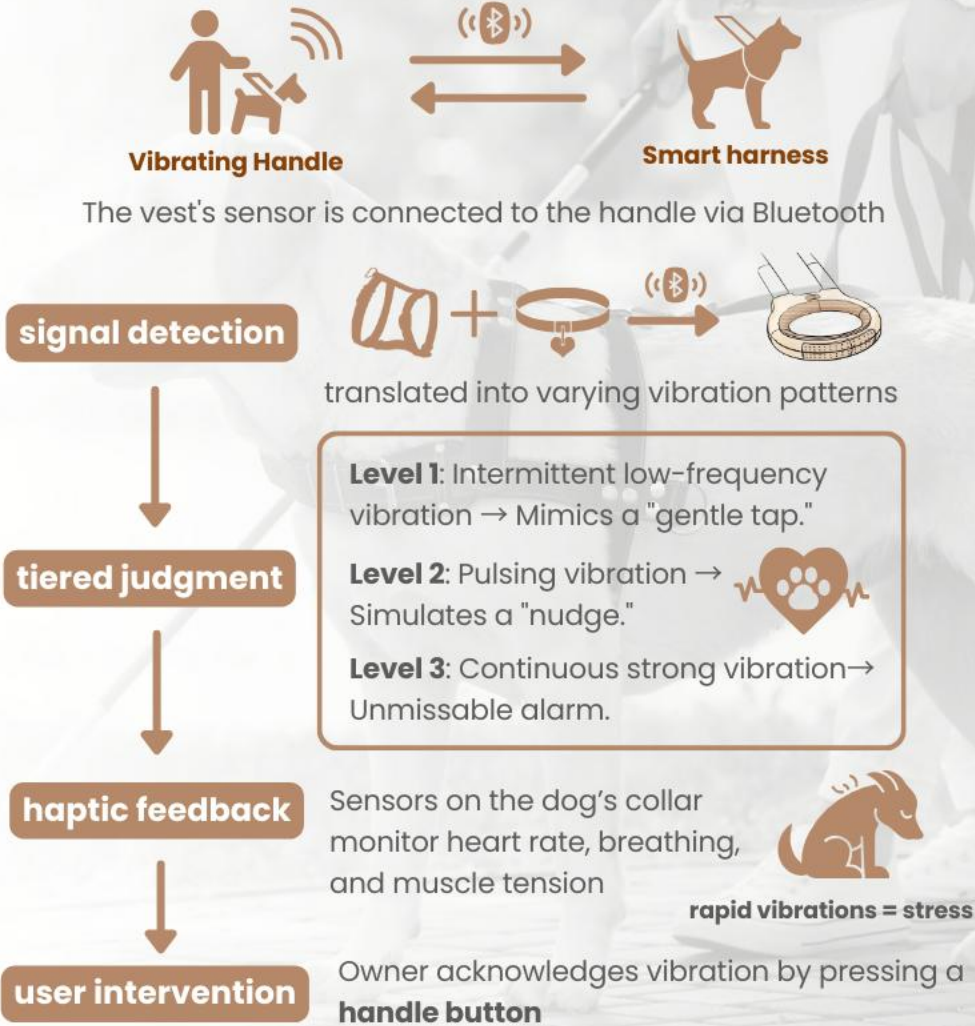
Guide dogs exhibit stress through elevated heart rate, panting, or gait changes.



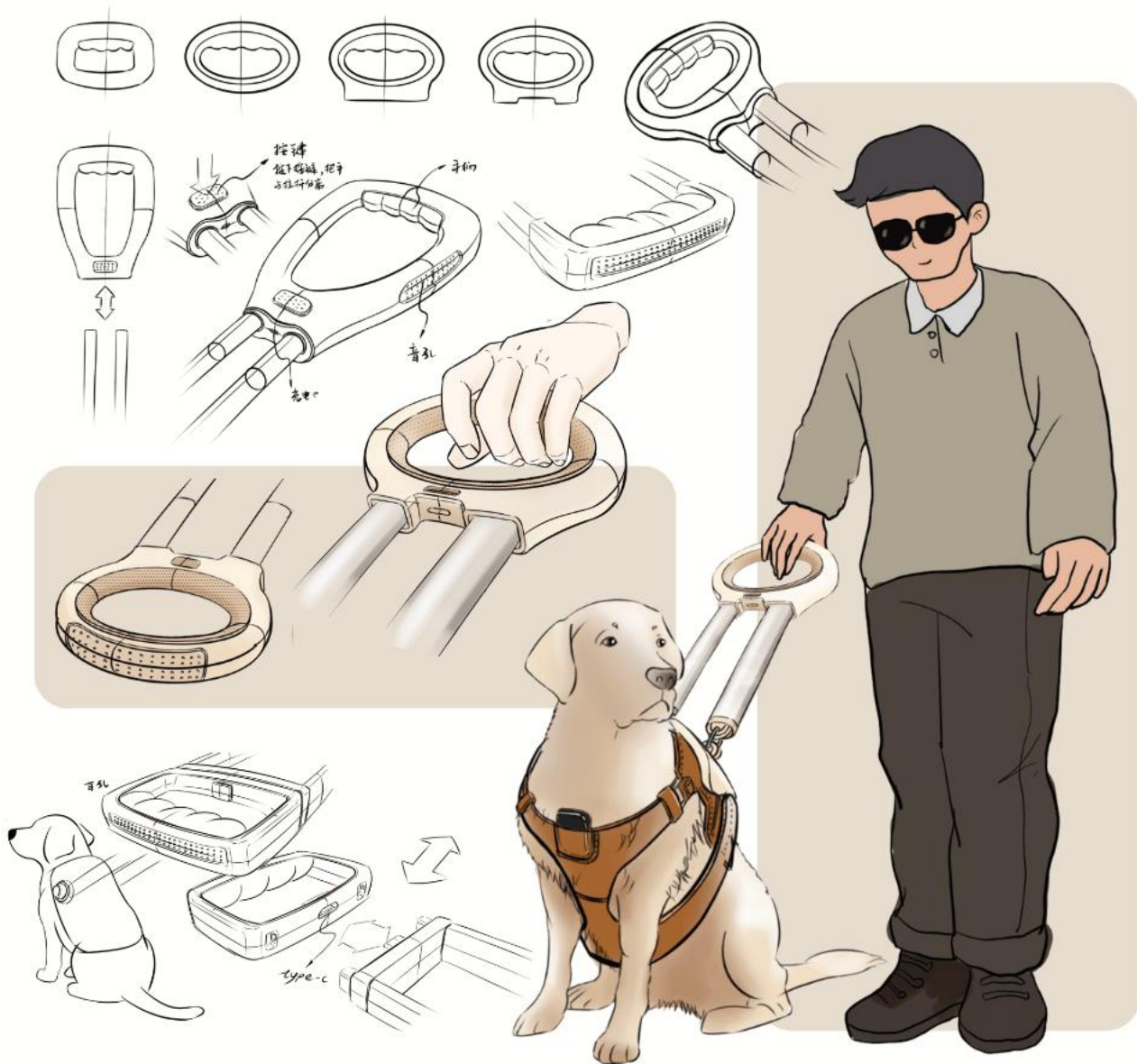
Blind users rely heavily on subtle cues (e.g., leash tension) to interpret their dog's state.

HMW

How might we establish bidirectional health synchronization between blind owners and guide dogs through a haptic feedback system?



IDEATION



PRODUCT MODELING

Braille handle for the blind



Dynamic Braille Display

Switch

Short Press :
power on/off

Long Press :
Bluetooth Pairing

Braille Message	Biological Signal	Sensor Trigger	Vibration
Happy	Normal	Regular status check	1s pulse
Tired	High	Body temp >39°C	5s pulses
Rest	Critical	Heart rate >140bpm	Continuous

The system alerts the handler immediately if the guide dog's resting heart rate consistently exceeds **140 BPM**—significantly above the normal range (**70–120 BPM**)—indicating potential stress, overheating, or cardiovascular risk.

Wearable technology for guide dogs



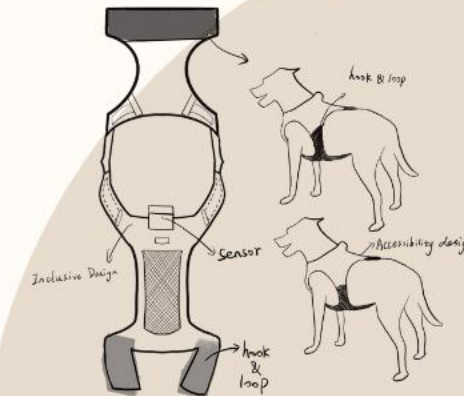
Heart rate sensor



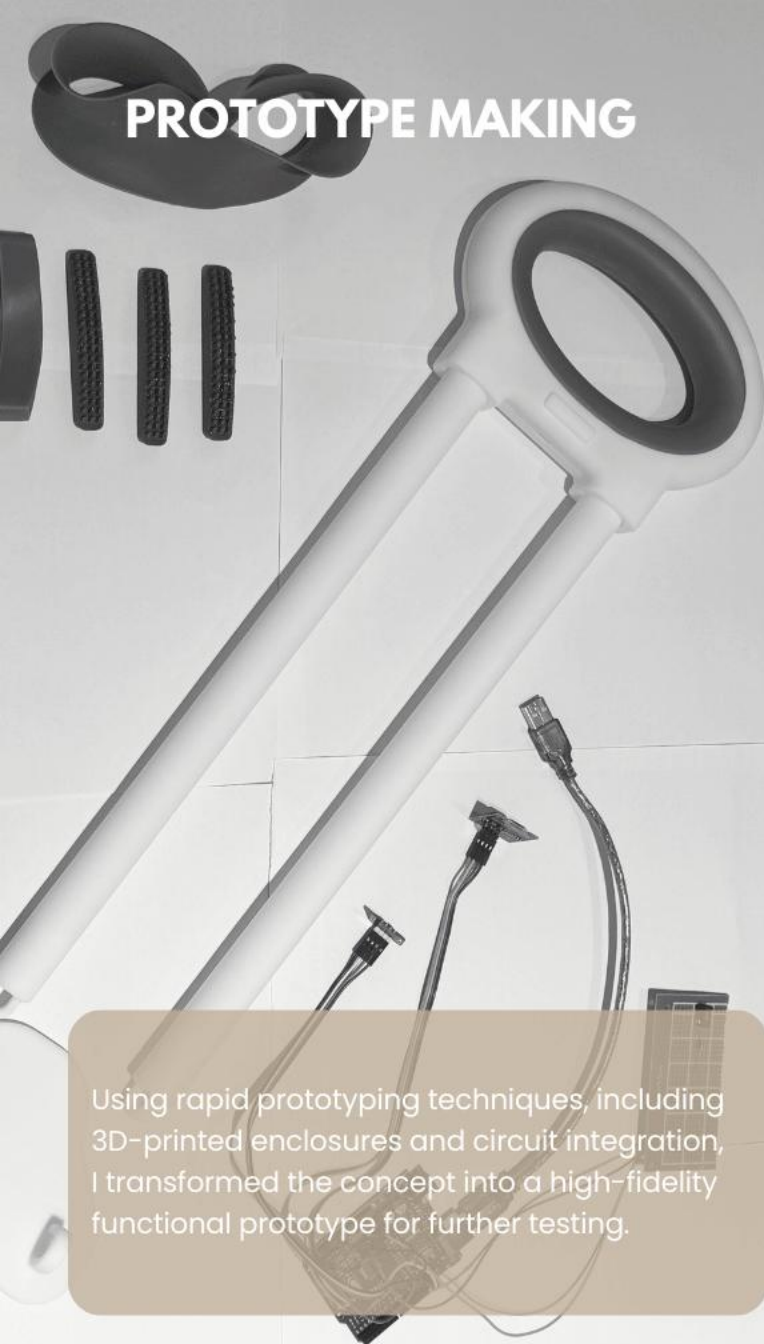
Hook and loop

Detect heart rate and temperature values

Convenience, large area easy to remove



PROTOTYPE MAKING



Using rapid prototyping techniques, including 3D-printed enclosures and circuit integration, I transformed the concept into a high-fidelity functional prototype for further testing.

Technique support



Making a guide dog harness



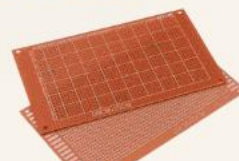
Prototype 3D printing



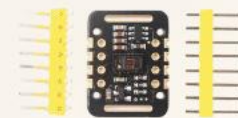
Braille module



Coin Vibration Motor



Universal board



MAX30102 sensor



0.96-inch SSD1306 OLED display

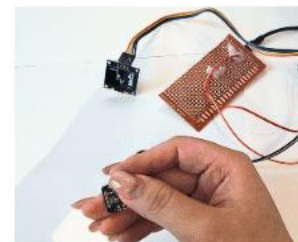
TECHNICAL RESEARCH & FEASIBILITY

Software & Algorithm Logic



Firmware Flow - Canine Side

1. Data Acquisition



fetches raw Photoplethysmogram (PPG) data.



Firmware Flow - Handler Side

1. BLE Listening



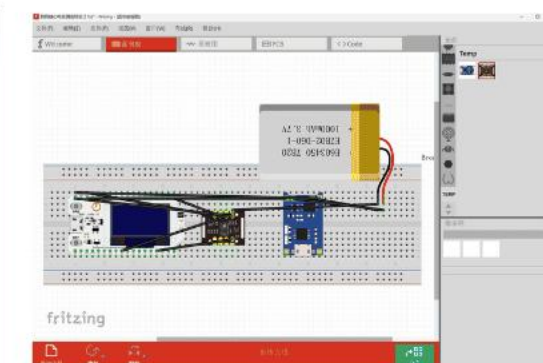
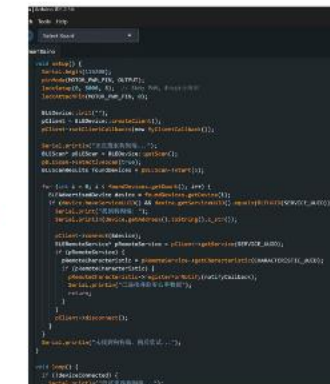
Continuously scan for specific Service/Characteristic from the canine device.



Main Loop

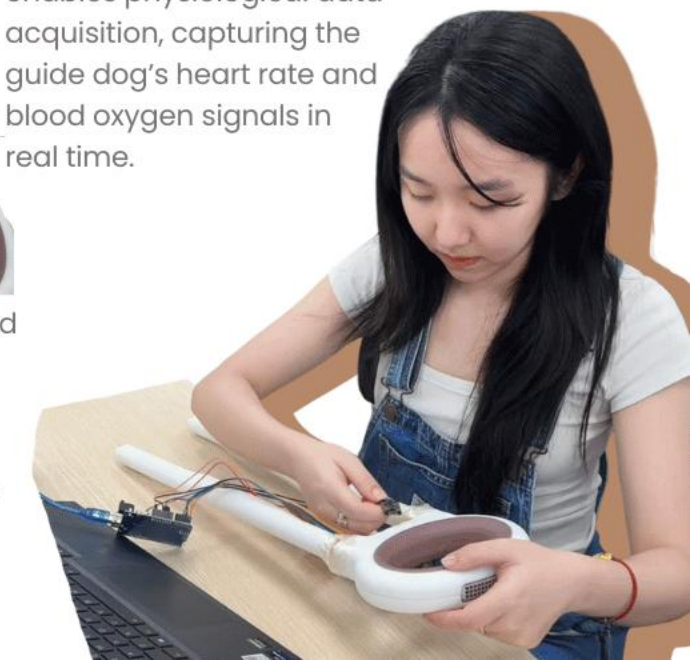
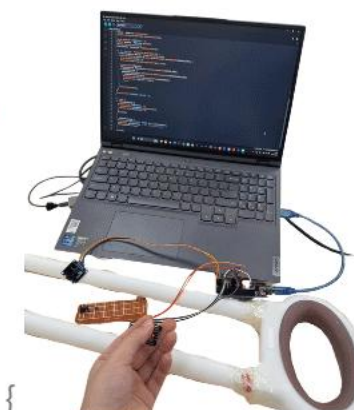
```
void onBLEReceived(value) {  
  int receivedHR = parseValue(value); // Parse received data  
  int vibrationIntensity = map(receivedHR, 140, 180, 50, 100); // Map heart rate to vibration intensity  
  analogWrite(MOTOR_PIN, vibrationIntensity); // Drive motor  
  activateBrailleAlert(); // Trigger Braille alert  
}
```

Coding



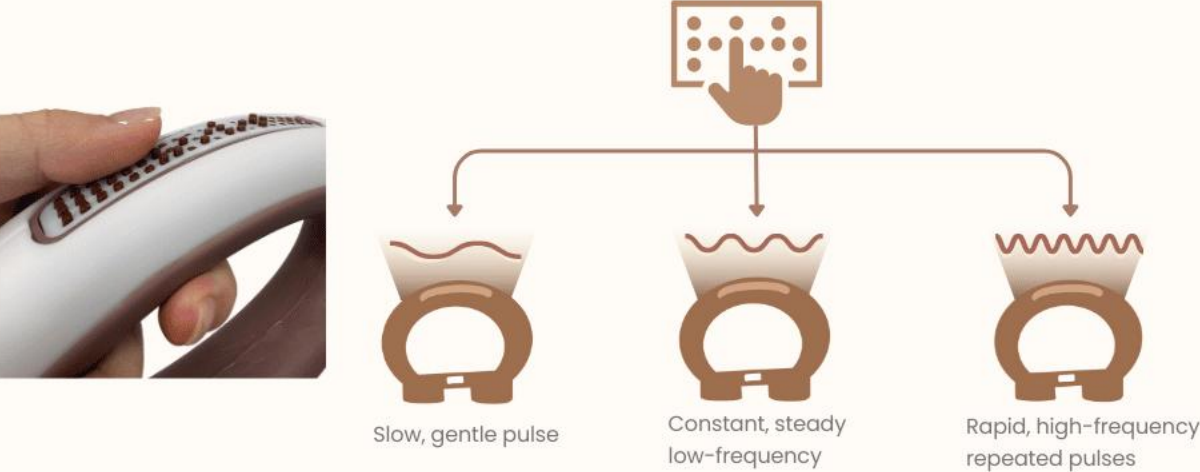
2. Data Parsing & Decision Making

The MAX30102 sensor enables physiological data acquisition, capturing the guide dog's heart rate and blood oxygen signals in real time.

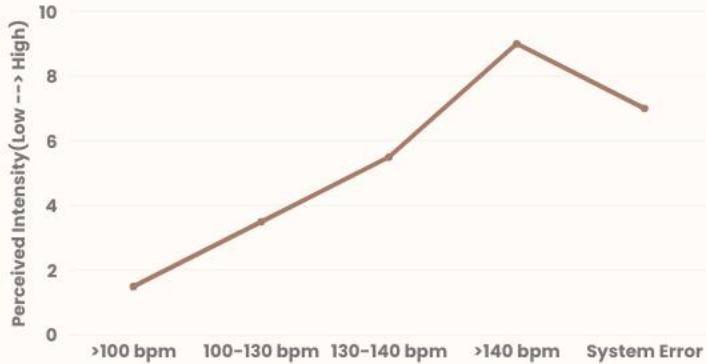


HAPTIC FEEDBACK HIERARCHY

The project designs a clear set of "haptic vocabulary" for your system, composed of different **Vibration Patterns**, **Intensities**, and **Rhythms**.



Haptic Feedback Intensity & Pattern Mapping

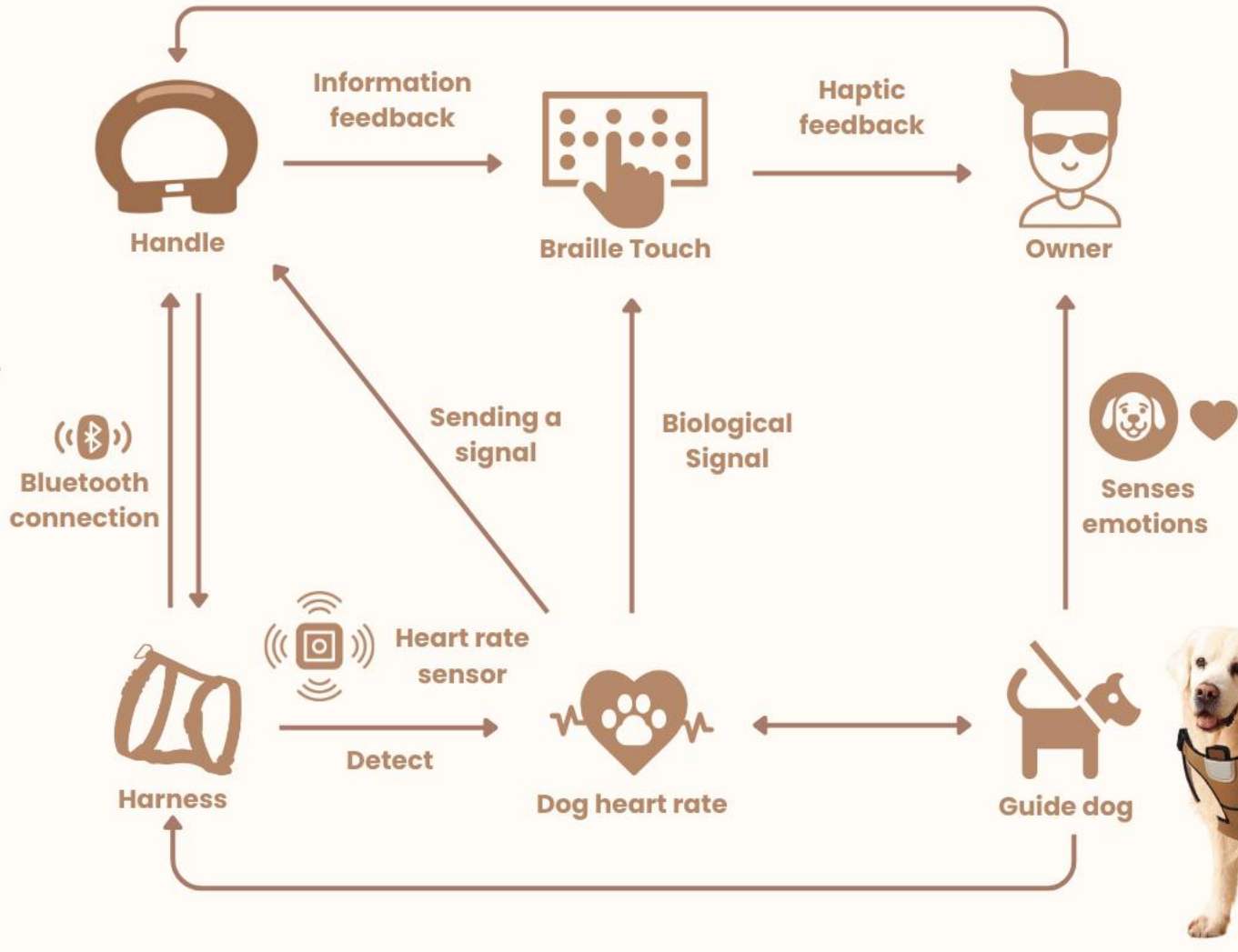


Through experience prototyping, we were able to refine the entire interaction flow, from the moment the alert is triggered to the handler's calming response to the dog.



WORKING FLOW

This diagram visually conveys my "systems thinking," illustrating data flow, component relationships, and user interactions.



“Your guide dog is sending you emotional signals!”

